

Introduction

This document describes the design necessities for the “All in One”, a project aimed at aiding Dr.Paz with organizing his classroom tools, improving safety when retrieving them, and storage. The goal is to do all the former three by improving the already existing storage cart Dr. Paz uses.

- 1. High priority- Adding a back board that reduces the angle at which the shelf tilts.**
 - a. It must reduce the wobble that happens when you move the board**
 - b. Shelf must be sturdy and stable for the entire day**
 - c. Dr. Paz agrees on the importance of the backboard.**
- 2. High priority-Adding 29 peg holders for glue guns**
 - a. Need to sturdy enough to hold glue guns for a year**
 - b. Needs to hold 29 glue guns**
 - c. Preferably 3D printed and modular.**
 - d. Dr. Paz agrees on the importance of the peg holders**
- 3. High priority-Add 9 drawers**
 - a. Each drawer should be able to hold enough material for each class**
 - b. Should be able to slide in and out**
 - c. Dr. Paz agrees on the importance of the drawers.**
- 4. Drawers should be clearly labeled and itemized (High Priority)**
 - a. Each drawer should say what it contains on it**
 - b. 2-3 drawers should be able to hold at least 2 different items**
 - c. Dr. Paz agrees on the importance of the labels.**
- 5. A compartment on the side to hold boxcutters and scissors (amount TBD, High Priority)**
 - a. Should safely hold all blades pointing into the holder**
 - b. Should slant at an angle allowing for more convenient retrieval**
 - c. Dr. Paz agrees on the importance of this compartment.**
- 6. Mobility**
 - a. Wheels need to work/roll much better**
 - b. Replace current wheels if needed**

- c. Dr. Paz agrees that these wheels should be high priority, as mobility is important to him.**

7. Needs to look aesthetically pleasing (Low Priority)

- a. Nice art on the side**
- b. Engineer themed, with tools and hard hats**
- c. Honestly Trivial, only worry about this if you've hit every deadline**